The Automation Paradox: Efficiency versus Resilience in Port Economics

Herong Bai, Chiyao Chen,
Yuheng Zhou, Lecheng Zhao

1. Introduction & Research Question

1.1 Case Hook & Economic Problem

In recent years, automation systems have been increasingly rolled out by major container ports as a means of boosting efficiency, reliability, and handling capacity. This technological shift has reshaped the day-to-day decision-making of port authorities, terminal operators, and shipping alliances alike. Ports are now confronted with a critical strategic dilemma: to commit to expensive investments in automation systems to maintain competitiveness, or to risk losing cargo traffic to more technologically sophisticated port hubs. For their part, shipping alliances, in turn, reallocate routes toward ports that offer higher reliability and faster turnaround, reinforcing these choices through network effects.

However, this transition has generated uneven outcomes across ports. Automation does not diffuse uniformly because ports differ in their ability to adjust capital and labour. Automated terminals require substantial upfront investment and long payback periods, making adoption risky and largely irreversible. At the same time, automation changes the skill composition of dock labour, replacing routine manual tasks with non-routine monitoring, coordination, and technical roles. Workers cannot adjust instantly to these new requirements, and ports vary widely in their capacity to retrain or redeploy labour.

As a result, only a subset of ports can upgrade smoothly, while others face rising relative costs and declining competitiveness. What initially appears as a technological upgrade, therefore, becomes an economic problem rooted in structural adjustment constraints rather than in technology itself.

1.2 Precise Research Question

This uneven adoption creates a broader coordination problem with system-wide consequences. When individual ports and firms make rational decisions based on private costs and constraints, shipping traffic becomes increasingly concentrated in a small number of automated hubs. While this improves efficiency at the node level, it reduces diversification in the global shipping network and increases exposure to disruptions such as labour disputes, cyberattacks, or technical failures.

The result raises the central research question of this study: How do structural rigidities in capital and labour adjustment under port automation generate asymmetric adoption, induce traffic concentration, and ultimately create a trade-off between efficiency and supply-chain resilience?

1.3 Thesis Statement

This essay argues that the core economic problem is not automation itself, nor unemployment in isolation, but structural rigidity in factor markets under technological change. High fixed investment costs and labour adjustment frictions amplify inequality in ports' ability to adopt automation, leading to self-reinforcing concentration through shipping alliances' routing decisions. These decentralised, privately optimal choices generate negative externalities at the network level, producing a resilience trade-off in global shipping. By analysing this mechanism through a detailed port-level case study, the paper demonstrates how efficiency gains at individual nodes can translate into systemic vulnerability when coordination and adjustment constraints are ignored.

2. Analytical Framework

2.1 Key Concepts & Model

This analysis is built on three core economic concepts: structural rigidities in factor adjustment, externalities, and coordination failure. Together, these concepts explain why port automation generates uneven outcomes across ports and leads to systemic supply-chain risk despite improving local efficiency. The primary analytical lens is institutional and industrial organisation

analysis, focusing on how firms and ports make decentralised investment decisions under constraints and how these decisions aggregate into network-level outcomes. Rather than assuming frictionless adjustment, this framework emphasises that capital and labour markets adjust slowly and asymmetrically under technological change.

First, structural rigidities refer to the limited ability of ports to adjust capital and labour inputs smoothly. Automation requires large, sunk investments in digital and mechanical systems and simultaneous changes in labour composition. These adjustments are costly, irreversible, and depend on local institutions such as financing capacity, labour laws, and training systems. As a result, ports face very different effective costs of adoption even when they have access to the same technology. Second, externalities arise because the benefits and risks of automation are not fully internalised by individual decision-makers. When a port automates, it improves reliability and attracts shipping traffic, but it also increases network dependence on that node. The resilience cost of concentration is borne by the entire supply chain rather than by any single port. Third, these features generate a coordination failure. Each port and shipping alliance makes rational decisions based on private incentives—minimising costs, reducing delays, and improving reliability—yet the collective outcome may be suboptimal due to increased systemic fragility.

2.2 Logic of application

This framework is applied to analyse how automation reshapes port competition and network structure through a clear causal chain.

We begin by examining how high fixed costs and labour adjustment frictions create asymmetric automation adoption across ports. Ports that can absorb capital costs and manage workforce transitions upgrade successfully, while others delay adoption or exit competitive routes. Next, we apply the framework to shipping alliances' routing behaviour. Alliances reallocate traffic toward high-performing automated ports to minimise operational risk. This behaviour amplifies initial performance gaps, leading to self-reinforcing concentration and endogenous market power, even in the absence of deliberate exclusion. Finally, we extend the analysis to the network level, where negative externalities emerge. While automation improves efficiency at the node level, concentration reduces diversification and increases vulnerability to disruptions such as cyber failures or technical breakdowns. The result is a resilience trade-off, in which decentralised optimisation leads to higher systemic risk.
This analytical framework allows the case study to move beyond description and test whether the observed data support the proposed causal mechanism linking structural rigidities, concentration, and resilience outcomes.

3. Case Analysis

3.1 Mechanism Diagnosis: How Automation Generates Concentration

This section examines how automation investment at the port level leads to traffic concentration and, ultimately, to changes in system resilience. Rather than treating automation as a simple efficiency upgrade, the analysis focuses on the mechanisms through which decentralised decisions accumulate into system-level outcomes.

Following Case Studies and Theory Development in the Social Sciences (George & Bennett, 2005), the case is used not to illustrate an outcome that is already known, but to uncover the sequence of causal steps that connect investment choices, routing behaviour, and network structure. This approach is especially relevant in infrastructure systems, where outcomes rarely emerge from a single decision and instead develop through gradual, reinforcing adjustments over time.

3.1.1 Identifying the Mechanism: A Process-Tracing Approach

To reconstruct this sequence, the study uses process tracing as its main analytical tool. Process tracing is useful here because it allows the analysis to move inside the case and examine how one decision alters the incentives and constraints faced by the next actor. As

emphasised by Process Tracing: From Metaphor to Analytic Tool (Bennett & Checkel, 2014), mechanism-based explanations depend on identifying observable intermediate steps, not just the correlation between inputs and outcomes.

In the port context, automation sets off a chain of adjustments rather than an immediate system-wide change. Large fixed investments and long payback periods mean that only a small number of ports can adopt automation early and continue upgrading. Once these ports achieve higher reliability and lower turnaround time variance, shipping alliances begin to shift routes toward them. These routing decisions are made independently, but they are not neutral: each additional service routed through an automated hub increases its operational attractiveness, making subsequent reallocations more likely.

The analysis, therefore, traces how early adoption reshapes routing incentives and how these incentives gradually reconfigure the network. Evidence is evaluated by checking whether changes in investment, routing, and throughput follow the expected order implied by the mechanism, consistent with the logic of within-case causal inference (Mahoney, 2012).

3.1.2 Concentration as a Network Externality

The resulting traffic concentration can be understood as a form of network externality, in which the value of using a particular port increases as more users choose the same node. This idea follows the classic formulation of network effects by network externalities, competition, and compatibility, but is adapted here to the context of physical infrastructure rather than consumer technologies (Katz & Shapiro, 1985).

Automation intensifies this dynamic because reliability itself becomes scale-dependent. As traffic grows, automated ports are better able to smooth operations and absorb variability, reinforcing their advantage over less automated terminals. What begins as a technical upgrade thus evolves into a self-reinforcing pattern of concentration, even though no single actor intends to create it.

3.1.3 Implications for System Resilience

While this process improves efficiency at the node level, it also alters the structure of risk. As more traffic is routed through a small number of hubs, the system becomes increasingly sensitive to disruptions at those locations. Delays or failures at automated ports, therefore, have wider consequences than similar shocks in a more distributed network. The outcome reflects a classic resilience trade-off: gains in efficiency are achieved at the cost of reduced redundancy, consistent with economic definitions of resilience as the capacity to absorb shocks without large systemic losses (Rose, 2004).

By tracing how automation changes routing behaviour and network structure step by step, this section shows that concentration is not an accidental by-product of technology but the predictable result of interacting incentives under capital rigidity and network effects. This diagnosis provides the basis for the evidence integration in the next section and for the policy discussion that follows.

3.2 Evidence Integration: Testing the Mechanism in the Shanghai Port Case

This section integrates case evidence to evaluate whether the causal mechanisms identified in Section 3.1 are observable in the development of Shanghai Port. Rather than treating the port as a success story of automation, the case is used to test whether capital rigidity, routing responses, and network effects unfold in the predicted sequence.

The analysis draws on public disclosures from Shanghai International Port Group (SIPG), throughput statistics, and comparative data from other major Asian ports. While these sources do not capture internal firm-level decisions, they are sufficient to examine whether the timing and direction of observable changes are consistent with the proposed mechanism.

3.2.1 Asymmetric Adoption: Capital Rigidity and Uneven Automation Upgrading

The first step of the mechanism predicts that automation adoption will be uneven across

ports due to high fixed costs and long payback periods. Evidence from Shanghai supports this pattern.

Shanghai began large-scale automation earlier and at greater intensity than many comparable ports, with continuous investment in automated terminals and digital logistics systems over multiple years. In contrast, several regional competitors, including Busan, only initiated full automated terminal development much later, with major projects beginning around 2023. This timing gap reflects differences in ports' ability to absorb large sunk investments and manage long adjustment horizons.

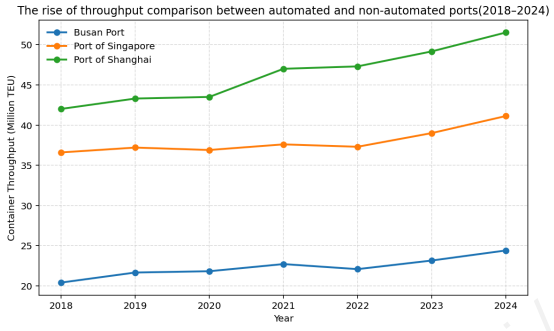


Figure 1

Figure 1 compares throughput growth between Shanghai and selected peer ports over time. While all ports experienced growth, Shanghai's throughput increased more rapidly after the expansion of automated capacity. This divergence is consistent with the prediction that early adopters benefit from cumulative upgrading, while late adopters face increasing difficulty catching up once automation becomes the industry benchmark. This pattern suggests that automation does not diffuse smoothly. Instead, capital rigidity creates a first-mover advantage that locks in early differences and generates asymmetric adoption trajectories across ports.

3.2.2 Route Reallocation and Traffic Concentration

The second step of the mechanism predicts that shipping alliances will reallocate routes toward automated ports to reduce operational risk. Evidence from routing patterns and throughput shares supports this expectation.

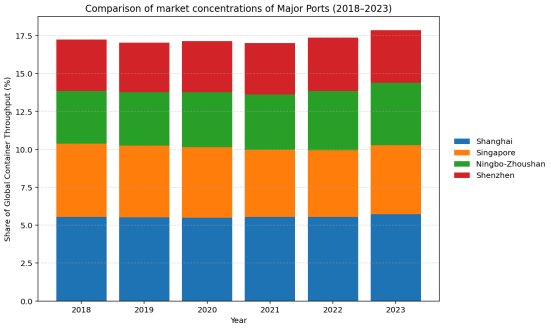


Figure 2

As Shanghai's automated capacity expanded, its share of national and global container throughput increased. Figure 2 shows that the proportion of total traffic handled by the largest ports has risen over time, indicating growing concentration. This trend cannot be fully explained by demand growth alone, as smaller and less automated ports did not experience similar gains.

The timing of these shifts matters. Increases in Shanghai's throughput follow automation upgrades rather than precede them, suggesting that routing decisions respond to reliability improvements rather than simply to size. Once traffic begins to concentrate, network effects reinforce the process: additional services routed through Shanghai further increase its attractiveness, encouraging even more reallocation. This evidence supports the interpretation of concentration as an endogenous outcome of decentralised efficiency-seeking decisions rather than a deliberate exclusion strategy.

3.2.3 Efficiency Gains and the Resilience Trade-Off

The final step of the mechanism concerns the system-level implications of concentration. While automation improves efficiency at individual nodes, it also increases dependence on a small number of hubs.



Figure 3: The externality of a rise in node efficiencies

As Shanghai's role in the network expands, disruptions at the port would have wider consequences for the entire supply chain than in a more distributed system. Although large-scale disruptions are rare, the structure of

the network itself implies higher systemic exposure. Figure 3 illustrates this trade-off conceptually: as efficiency rises with automation, redundancy declines, increasing vulnerability to localised shocks. This pattern reflects a classic resilience trade-off. Individual actors optimise locally by choosing the most reliable ports, but the resulting network becomes less diversified. The external cost of concentration is therefore not internalised by any single decision-maker, reinforcing the need for system-level policy responses.

3.2.4 Summary: Evidence Consistent with the Proposed Mechanism

Taken together, the Shanghai case provides within-case evidence consistent with the mechanism identified in Section 3.1. Capital rigidity produces uneven adoption; routing responses amplify early advantages; and network effects transform local efficiency gains into system-level fragility. While the analysis relies on public data and cannot observe internal alliance contracts, the timing and direction of observable changes align closely with the predicted causal sequence.

This strengthens the argument that concentration and resilience risk are not accidental outcomes but the predictable result of decentralised decision-making under structural constraints.

4. Implications & Proposals

4.1 Targeted Proposals

The following proposals will address the root cause of the structural rigidities in factor adjustment from three angles: the labour side, the capital side, and the over-concentration of markets.

The first proposal is a cooperative project between the government and dock firms to mitigate the negative effects of labour structural constraints. Firms would be responsible for introducing new non-routine dock jobs to replace conventional physical ones, in which manual workers previously had a comparative advantage. These new jobs should comprise a wide variety of roles. There should be lower-skilled positions to facilitate the transition of dockworkers who previously performed more basic tasks, including simple monitoring for documentation or safety inspections, as well as roles requiring basic communication skills, such as scheduling and coordination. On the other hand, as automated ports require technological support and protection, higher-skilled jobs should also be introduced, including cybersecurity support to prevent cyberattacks, digital monitoring, and technical maintenance. Simultaneously, governments should reform the outdated dock laws, which include legally recognising new dock jobs and guaranteeing reemployment in port-related activities within or between different docks and ports.

Subsequently, governments should establish retraining programs on a rolling basis to support dock firms. Port businesses would identify and select workers at high risk of substitutability and send them to these retraining programs. Governments would recruit specialists to train dockworkers during this period. These specialists would collaborate to jointly determine a set of criteria for assessing qualifications. Since workers have different learning speeds, those who pass the assessment early could return to work sooner. Once the majority of workers from the first round of training return to their jobs, firms would send the next group of workers with the second-highest risk of substitutability. Additionally, because dockworkers in the retraining program would not receive wages, governments would need to provide adjustment funds to sustain their livelihoods.

The second proposal is another cooperative plan, this time between the government and firms that develop and manufacture automated machines. The purpose of this plan is to alleviate capital rigidity by facilitating the leasing of automated equipment, thereby increasing flexibility and reducing sunk costs. Initially, the government should establish legal frameworks for leasing automated machines, including rules on ownership and liability, while guaranteeing protection and the right to appeal to the courts and seek compensation for both parties. The framework should also introduce different contractual types, such as long-term leasing, pay-per-use, or payment based on total throughput. Subsequently, the government should standardise the process by

unifying operational and technical standards across different ports, such as electricity requirements. This step is critical for disseminating the system to multiple ports and docks nationwide. Finally, a platform or system should be created to connect automated machine producers with dock firms, functioning like a bridge for coordination and ordering.

The third proposal requires governments to nationalise certain port infrastructure and standardise key operations and automation mechanisms to prevent asymmetric adoption and market concentration, thereby addressing negative externalities. Governments should nationalise key features of ports that constitute natural monopolies, defined as markets where a single firm can serve at a lower cost than multiple firms while fully exploiting economies of scale (Khemani et al., 1990). These features may include terminal operating systems, traffic management systems, and power supplies. In addition, governments should standardise processes such as automation systems, data protocols, and operating procedures to ensure that ports in different regions have similar access to automation technologies. Together, these measures reduce technological fragmentation and limit excessive market domination by highly automated ports, while still preserving healthy competition among private firms in service provision, ensuring overall efficiency.

4.2 Brief Feasibility & trade-offs

The greatest trade-off for the first proposal is the substantial government funding required to establish these new training programs. This could reduce the budget available for other important areas, such as infrastructure, and may result in higher taxes for citizens. For the second proposal, the main drawback is that while leasing reduces rigidity and fixed costs, it could generate increased expenses elsewhere, particularly high transportation costs for large, heavy automated machines. Finally, the third proposal has two main limitations. The first is the potential inefficiency of government-managed infrastructure, as there is little pressure for survival. The second is that the boundaries of control over different aspects of ports may be unclear, which could lead to confusion and conflict between private firms and the government.

5. Limitation and Conclusion

5.1 Concise Conclusion

In conclusion, this essay argues that the economic problems resulting from port automation are fundamentally driven by structural rigidities in factor adjustment. These factors include mainly labour and capital involved in the transformation. Additionally, the technological optimisation of individual firms may increase overall systemic risk, resulting in asymmetric adoptions and market concentrations. Consequently, other firms may face resilience trade-offs, resulting in negative externalities. To address these issues, this essay proposes three measures: a rolling retraining programme alongside general labour market reforms, the leasing of automated machinery, and, lastly, the nationalisation and standardisation of key infrastructures and mechanisms of ports.

5.2 Key Limitations

The analysis conducted in this essay has three key limitations: the sole reliance on public data, the use of an extreme case, and the long-term dynamic nature of automation. The reliance on public data, as it is not possible to access internal metrics on cost structures and labour movements, does not allow for an accurate measurement of the exact effect size of automation. Additionally, the use of an extreme case—Shanghai Port in this context—means that the results or observed trends may not apply to smaller ports. Furthermore, the analysis of the economic effects of automation in this essay focuses on present and past data. As a result, these findings may be less relevant in the future, as automation is a dynamic and long-term process that is constantly evolving.

References

1. George, A. L., & Bennett, A. (2005). *Case Studies and Theory Development in the Social Sciences*. The MIT Press. http://bvbr.bib-bvb.de:8991/F?func=service&doc_library=BVB01&local_base=BVB01&doc_number=013153145&sequence=000002&line_number=0001&func_code=DB_RECORDS&service_type=MEDIA
2. Bennett, A., & Checkel, J. T. (2014). *Process tracing: From metaphor to analytic tool*. https://doi.org/10.1017/cbo9781139858472
3. Mahoney, J. (2012). The logic of process tracing tests in the social sciences. *Sociological Methods & Research*, *41*(4), 570–597. https://doi.org/10.1177/0049124112437709
4. Katz, M. L., & Shapiro, C. (1994). Systems competition and network effects. *The Journal of Economic Perspectives*, *8*(2), 93–115. https://doi.org/10.1257/jep.8.2.93
5. Katz, M. L., & Shapiro, C. (1985). Network externalities, competition, and compatibility. *American Economic Review*, *75*(3), 424–440. https://ci.nii.ac.jp/naid/10021349779
6. Rose, A. (2004). Defining and measuring economic resilience to disasters. *Disaster Prevention and Management an International Journal*, *13*(4), 307–314. https://doi.org/10.1108/09653560410556528
7. Khemani, R. S., Shapiro, D. M., & Organisation for Economic Co-operation and Development. (1990). GLOSSARY OF INDUSTRIAL ORGANISATION ECONOMICS AND COMPETITION LAW. In *GLOSSARY OF INDUSTRIAL ORGANISATION ECONOMICS AND COMPETITION LAW*. https://www.oecd.org/regreform/sectors/2376087.pdf

Herong Bai / Chiyao Chen / Yuheng Zhou / Lecheng Zhao herongbai.com